

Zijun Jiao

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RESEARCH INTERESTS

Data-driven optimization and decision-making for sustainable operations, with interests in demand and condition forecasting, robust and resource-aware predictive modelling, and the integration of predictive models with optimization for maintenance, capacity, and service decisions.

EDUCATION

- **Tsinghua University** Beijing, China
Research Assistant (non-degree appointment), The Future Laboratory Aug. 2025 – Present
- **National University of Singapore** Singapore
M.Sc. in Data Science for Sustainability Aug. 2024 – Jun. 2025
- **University of Southampton** Southampton, United Kingdom
B.Sc. (Hons) in Mathematical Sciences Sep. 2021 – Jun. 2024

RESEARCH EXPERIENCE

- **Backend and Tool-Agent Orchestration for an Air-Gapped Railway Intelligence System**
Tsinghua University, Research Assistant, advised by Prof. Haipeng Mi Jun. 2025 – Present
 - **Backend and integration:** Responsible for backend construction, covering data processing and the integration of external tool agents, within a network-isolated on-premise language-model system supporting office and maintenance workflows at a heavy-haul railway operator;
 - **Meeting-documentation workflow:** Designed and built the automated meeting-minutes workflow as a two-tier agent: a controller that selects among registered retrieval, transcription and reasoning tools, and a sub-agent that converts noisy transcripts into structured minutes through noise reduction, semantic reconstruction and topic structuring;
 - **Reliability and data governance:** Introduced a permissioned four-zone workspace (read-only knowledge base, write-isolated user input, scratch space, results directory) together with an explicit delegation contract between the two tiers, so that agent actions cannot modify authoritative enterprise records, which is the governance requirement motivating on-premise deployment.
- **Resource-Aware Benchmarking of Classical and Deep Time-Series Forecasters**
National University of Singapore, course research project Jan. – May 2025
 - **Study design:** Benchmarked six forecasters (Prophet, LightGBM, DLinear, LSTM, Transformer, DARNN) on five univariate series drawn from energy consumption, air quality, manufacturing productivity, urban traffic and biomedical gait, each with a dataset-specific preprocessing pipeline covering imputation, winsorisation, calendar and event features, and train-set-only scaling to prevent leakage;
 - **Evaluation beyond error:** Scored every model and dataset pair on four axes: forecast error (MAE, RMSE, MAPE, sMAPE), decision-relevant accuracy (directional and threshold accuracy), deployment cost (training time, peak memory, early stopping), and noise robustness (Δ MAE under additive Gaussian perturbation, $\sigma = 0.05$);
 - **Finding:** Peak memory varied by roughly three orders of magnitude across models, from tens of gigabytes for the attention-based recurrent model to tens of megabytes for the linear and gradient-boosting baselines, and the most accurate model on a given series was frequently neither the cheapest nor the most noise-stable. LightGBM degraded by up to 63% MAE under perturbation on the smallest series while the recurrent models stayed within 1.2%, and several deep models fell below 0.50 directional accuracy on that series. The study concludes that forecaster selection should be posed against deployment constraints rather than as error minimisation alone.
- **LLM-Assisted ESG Metric Extraction and Standardisation for Healthcare Reporting**
National University of Singapore 2025
 - **Extraction and standardisation architecture:** Paired a language-model extractor, which parses heterogeneous sustainability-report prose into metric, value, unit and confidence tuples, with a BERT semantic matcher mapping free-form metric names onto a canonical taxonomy, connected by typed data contracts so that extraction, matching and scoring remain independently replaceable;

- **Scoring framework and validation plan:** Implemented an 18-topic, 39-indicator rubric aligned to SGX reporting guidance, with topic weights derived from mention frequencies across 83 Singapore-listed healthcare sustainability reports, and designed an evaluation protocol for human annotation and external-rating comparison to be carried out in a subsequent validation phase.
- **Machine-Learning Representations for Quantum Many-Body Ground States**
Undergraduate dissertation, sole author; advised by Dr. Itamar Yaakov *Oct. 2023 – May 2024*
 - **Independent theoretical dissertation:** Reviewed and compared neural-network quantum states, matrix-product and projected entangled-pair tensor networks, and restricted Boltzmann machines as compressed variational representations of exponentially large wavefunctions, covering variational optimisation, imaginary-time evolution, and variational Monte Carlo with stochastic reconfiguration;
 - **Comparative analysis:** Analysed the expressivity relationship between restricted Boltzmann machine and matrix-product-state representations and the conditions for conversion between them, using entanglement entropy as the comparison criterion.

COURSEWORK PROJECTS

- **Multi-Agent Equity Analysis with Large Language Models**
National University of Singapore *2025*
 - **Shared-state coordination:** Co-designed the shared state object through which four analyst agents (fundamentals, technical, sentiment and adversarial) exchange information. Agents never call one another directly: data-collection tools write into six typed domains covering identifiers, market data, financial statements, external research, generated reports and workflow control, and downstream agents read the fields they need, which keeps each agent independently testable and replaceable;
 - **Two-tier prompt architecture:** Separated a system-level prompt fixing output format, tool-use protocol and fallback behaviour from role-specific prompts constraining each analyst's methodology and risk boundaries, so that agent outputs remain machine-parsable and mergeable across roles;
 - **Adversarial review stage:** Specified a five-part challenge applied to the three analyst reports, testing for over-optimistic assumptions, under-weighted macro and governance risk, cross-report contradictions, assumption fragility, and horizon-separated forecasts, as a hedge against convergent reasoning among the analyst agents. The project delivered a working system; no backtesting or forecast-validity evaluation was carried out.
- **Scaled Conjugate Gradient Methods for Deep Learning Optimisation**
National University of Singapore *2025*
 - **Study context:** The team studied a hybrid optimiser introducing a dynamic scaling factor into the conjugate-gradient update rule, with convergence rates argued at $O(1/N)$ under a constant step size and $O(1/\sqrt{N})$ under a decaying step size, benchmarked against SGD, Adam and AdaBelief on CIFAR-10, CIFAR-100 and IMDB classification;
 - **Individual contribution:** Wrote the discussion and possible-extensions analysis, examining which assumptions in the convergence argument constrain the method's applicability to non-convex settings and which extensions are worth pursuing. The derivation and the empirical work were carried out by other team members.

RELEVANT COURSEWORK

- **Optimisation and numerical methods:** Optimisation Methods for Deep Learning; Numerical Methods
- **Probability, statistics and modelling:** Probability; Statistical Modelling; Deep Learning for Time-Series Analysis
- **Mathematics:** Partial Differential Equations; Real Analysis; Linear Algebra; Group Theory

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, L^AT_EX
- **Modelling:** PyTorch, Darts, LightGBM, Prophet, scikit-learn, Hugging Face Transformers, NumPy, Pandas
- **Systems:** LangChain, agent and tool orchestration, retrieval-augmented generation, MySQL, Flask/Dash, Git